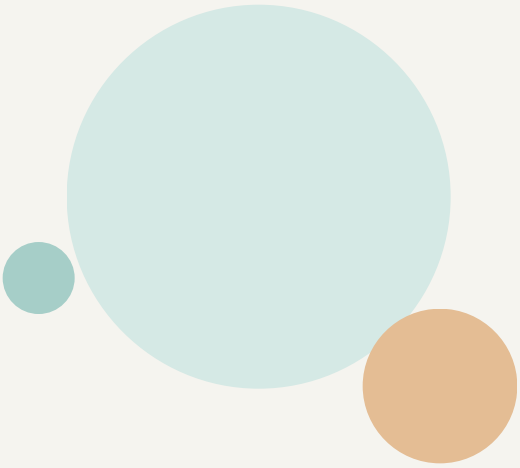
TOPIC MAP v3
24 OCT 2026

F01



COMPLETE LEARNING GUIDE

Business Framing, Requirements & Quality Attributes

A beginner-to-senior reference for turning ambiguous business problems into measurable, defensible architecture decisions.

17 PROGRESSIVE LESSONS

05 CORE CHECKLIST OUTCOMES

01 SENIOR DECISION CHALLENGE

READER'S MAP

Learn the business problem before choosing technology.

This guide follows the authoritative F01 syllabus. Work from basic terms to practical decisions; complete exercises before using the separate answer key. The five required items are C014–C018.

01	02	03	04
Understand	**Discover**	**Measure**	**Defend**
Users, roles and journeys	Stakeholders and requirements	Quality and business outcomes	Senior trade-offs and ownership

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> **Reading rule:** Lessons L01–L17 first; independent practice next; answer key afterward. A completed reading is not evidence of mastery.

Complete progressive reference guide | Topic map v3 | Prepared 2026-10-24
Status: Reference document, not a live lesson or a completed assessment.
Target: Beginner foundations → senior Solution Architect interview reasoning.
Scope: F01 only: **C014, C015, C016, C017, C018**, all **Core**.
Authority: `solution_architect_interview_topic_map_v3.md` , section **F01** and its learning/assessment rules.

Important ID correction — Source requirement: The request names `B01 (C014–C018)`. In the authoritative map, **B01 has C001–C004** (software/Python basics), whereas **F01 has C014–C018**. This guide follows **F01/C014–C018** without adding B01's software curriculum. No syllabus IDs have been reassigned.

How to use this reference

Read lessons **L01 → L17** in order. For each lesson, use the sequence **understand → explain → demonstrate → practice → evaluate → memorize**. Work through its example, finish its exercise without looking at the solution, use the answer key only afterward, then answer its interview follow-up aloud. A lesson is **not passed** simply because you read it. The goal is to explain, apply, and defend a trade-off **without notes**. The cumulative brief and senior decision challenge come after the lessons.

Source labels used throughout:

- **[Source requirement]** = mandatory scope or teaching/assessment rule from topic map v3.
- **[Verified external fact]** = supported by an identified primary source `[R#]` in the references, checked **2026-10-24**.
- **[Hypothetical practice assumption]** = fictional actors, business rules, targets, budgets, measurements, or scenarios. Never treat them as American Express or industry commitments.
- **[Recommendation]** = a proposed method or professional judgment. Test against a real organization before adoption.

[Source requirement] The map's role label is *Sr Solutions Architect I, American Express Global Commercial Services*. The exact vacancy, company internal standards, interview format, and technology stack are **not independently verified**. This guide does not claim to describe them. The exercise is fictional; a public American Express page is optional domain context only [R7].

What is inside and outside F01?

Inside: discover the business problem and target outcome; distinguish customers and users; map journeys and stakeholders; elicit and classify requirements; specify and rank availability, latency, security, data freshness, and changeability; connect business key performance indicators to technical measures; defend trade-offs and owners.

Outside: implementation-level component diagrams and architecture decision records (F02); detailed commercial-payment rules (D01/D02); network/API implementation (B02/T03); detailed data models (B03/T05); security-control design (T06); production service-level-objective engineering (T07); advanced Python (T11). We use minimal terminology only where F01 needs it.

Prerequisite check — before L01

No prior business-analysis knowledge is assumed. You only need to understand these everyday words:

Word	Simple meaning	Example
Business	An organization providing products or services	A company buys office supplies
Customer	A person or organization receiving value	A company buys a spending-management service
Goal	Something the organization wants to achieve	Reduce manual expense reviews
Problem	Something stopping the goal	Transactions appear too late for finance
System	People, processes, and/or software working together	A spending-control service and the people using it
Data	Recorded facts	Purchase amount and event time
Measure	A number or observation used to judge results	Minutes until a purchase becomes visible
Trade-off	Gain one benefit by giving up some other benefit	Faster display can mean showing preliminary, not verified, information

Prerequisite questions (P0; self-check): (1) What is the difference between a business goal and a software feature? (2) Why can two users of one system have different needs? (3) Why is “make it fast” not enough to test a solution? Answers are in **AK-P0**.

Running case and scope of assumptions

[Source requirement] The map defines a hypothetical commercial platform in which business administrators set spending rules, employees initiate purchases, finance teams view and reconcile transactions, and the platform connects to identity, risk, payment, and reporting capabilities.

New words: A **spending rule** is a business limit on spending. **Identity** establishes which user is acting. **Risk checks** assess whether an activity requires more review. **Payment** here means the business capability involved in handling the purchase; its internal processing is deliberately unspecified. **Reporting** displays recorded transactions. **Reconciliation** means checking that two sets of relevant records agree and investigating differences. An **integration** is a connection between capabilities. No technical integration design is required in F01.

[Hypothetical practice assumptions] Our fictional client is **Northstar Office Group**, which has multiple employee teams. Its administrators can define a spending policy; employees request purchases; risk may require a check; finance sees a transaction record and investigates exceptions. The fictional starting complaint is **“Finance cannot see purchases soon enough.”** No baseline performance or official target has yet been confirmed. All proposed times, quantities, percentages, costs, and organization roles below are invented for practice and **not claimed as facts**.

Important distinction: A **record is visible** when the system displays it. **Verified** means it has passed the agreed checks. **Final** means the relevant business process has reached a final state; *visibility does not prove final payment or settlement*. The detailed financial lifecycle belongs to D02.



Module A — Understand the business and its users (C014)

F01 / L01 / CORE CURRICULUM

L01 — Customer types versus user roles

Checklist: C014 · **Assessment:** A01.

1. Purpose and prerequisites

Purpose: Know *who receives value* and *who does the work* before choosing a solution. Requires only the prerequisite definitions above.

2. Terms and fundamentals

- **Customer type** groups customers by a meaningful need or context: for example, a small organization and a multi-department enterprise. These are *segments*, not verified Amex categories.
- **User** is the person interacting with a solution. **User role** is what that person is allowed or expected to do in a particular task.
- **Persona** is a short, evidence-based description of a user type. A fictional persona can help practice, but do not mistake it for customer research.
- **Buyer**, **payer**, and **user** need not be the same person. The buyer chooses the product; the payer funds it; a user operates it.
- **Need** is the problem to solve. **Feature** is one possible capability provided by software. A feature does not automatically prove value.

[Verified external fact] The International Institute of Business Analysis (IIBA) defines end users and stakeholders separately, and distinguishes business needs from solution features [R1].

3. Worked example — step by step

Category	Fictional example	Need, not feature
Customer type: small firm	One office, one finance generalist	Fewer manual checks
Customer type: enterprise	Many departments and reviewers	Consistent spending governance across teams
Administrator role	Sets spending limits	Prevent unauthorized spending
Employee role	Requests a purchase	Know quickly whether to proceed
Finance role	Checks transaction records	Trust timely, correct information

Walkthrough: (1) First identify the paying organization as customer. (2) Divide customers only where the need changes. (3) List roles within that organization. (4) State one job and expected outcome for each role. (5) Do not assume every employee can change a limit: a *role* is not automatically a permission.

4. Guided exercise E01

Goal: Separate three customer types/roles without guessing permissions. **Steps:** For a university with several departments, write the *customer organization*, two possible user roles, each role's need, and one unresolved permission question. **Expected:** One customer type, two roles, two needs, one open question. **Hint:** "Who pays?" and "Who uses?" differ.

5. Failure, detection, recovery, trade-off

Failure: You call "finance" the customer and ignore employees. **Detection:** Employee needs vanish from acceptance discussions. **Recovery:** Map buyer and every operating role. **Trade-off:** More segmentation improves fit but makes discovery and support more complex. Segment when user needs materially differ.

6. Interview Phrasing

> "I separate the customer organization from the users and their roles. That helps me test whether the solution meets both the buyer's outcome and the day-to-day needs of administrators, employees, and finance."

Plain English: A company buys the solution; several people use it for different jobs. **Follow-up F01:** Can one person have two roles? What must you verify? **Answer check:** AK-E01/F01.

F01 / L02 / CORE CURRICULUM

L02 — Business capabilities, not technology boxes

Checklist: C014 · **Assessment:** A02.

1. Purpose and prerequisites

Purpose: Describe what the organization must be able to do without prematurely choosing software. Requires L01's customers and roles.

2. Terms and fundamentals

- A **business capability** is an ability the organization needs, such as *manage spending policies*. It answers **what**, not necessarily **how**.
- A **business process** is an ordered series of tasks: *request → check → decide*.
- A **feature** is a software behavior such as a "Change limit" button.
- A **capability map** groups related abilities; it is not a deployment diagram.

[Verified external fact] The IIBA glossary describes capabilities and business processes as different concepts [R1].

3. Worked example — step by step

```
Desired outcome: Control employee spending and reduce manual review
    |
    +-- Manage spending policies       [business capability]
    |     +-- define / update policy    [business activities]
    +-- Check purchase requests
    |     +-- apply policy to request
    +-- Show transaction information
    |     +-- label status and time
    +-- Investigate discrepancies
          +-- review unmatched records
```

(1) Start with a *business outcome*. (2) Name capabilities as verb+noun: “Manage policies.” (3) Name activities needed inside them. (4) Avoid implementation nouns such as “microservice,” “database cluster,” or “dashboard technology.” (5) Test each capability by asking which business outcome it supports.

4. Guided exercise E02

Goal: Produce a small capability map. **Steps:** From the Northstar case, identify four capabilities, one task per capability, and one user role helped by each. **Expected:** Four capabilities with traceable user value. **Hint:** The case mentions rules, purchases, records, and exceptions.

5. Failure, detection, recovery, trade-off

Failure: Treating “build a mobile application” as the business capability. **Detection:** The list says nothing about why customers need it. **Recovery:** Restate as “Submit purchase request from a device,” then ask if mobile access is necessary. **Trade-off:** Broad capability maps communicate strategy; detailed ones improve planning but risk becoming solution designs. Stop at the level needed to frame F01.

6. Interview Phrasing

> “I would describe the needed business capabilities first, such as policy management and transaction visibility. That keeps the discussion focused on outcomes before we decide how to implement them.”

Plain English: Describe the work before picking a tool. **Follow-up F02:** Could one capability be provided by several systems? **Answer:** AK-E02/F02.

F01 / L03 / CORE CURRICULUM

L03 — End-to-end customer and user journeys

Checklist: C014 · **Assessment:** A03.

1. Purpose and prerequisites

Purpose: Find missing steps, delays, handoffs, and exceptions. Requires L01 roles and L02 capabilities.

2. Terms and fundamentals

- A **journey** is the sequence of experiences and actions from a starting need to a meaningful result.
- **Trigger** starts the journey; **touchpoint** is a point of interaction; **handoff** moves work between roles/capabilities.
- **Happy path** is the expected successful route. **Exception path** handles failures or special cases.
- **End-to-end** means covering the whole business result, not just one software screen.
- A **pain point** is a step causing delay, confusion, cost, or risk.

3. Worked example — step by step

```
TRIGGER: employee needs office supplies
  1. Employee creates request
  2. Spending rule is checked
     +-- rule allows ------> 3. Purchase may proceed
     +-- rule blocks ------> 3a. Show reason / next action
     +-- review needed ---> 3b. Wait for authorized review
  4. Transaction status becomes available to finance
  5. Finance checks record and resolves any discrepancy
  END: purchase is handled and finance has usable information
```

(1) Name the trigger and the user. (2) Record what they do and what they expect at each step. (3) Show decisions and exceptions. (4) Mark the transition to finance. (5) Define what “done” means for each user. **No payment-processing internals are assumed.**

Step	Role	Desired result	Risk/pain point	Discovery question
Request	Employee	Clear next action	Unclear rejection	Is an explanation needed?
Rule check	Admin/policy owner	Consistent application	Wrong policy	Which policy version applies?
Status view	Finance	Reliable information	Stale or provisional data	What status and age may be shown?
Exception	Finance/risk	Safe resolution	Missing owner	Who decides escalation?

4. Guided exercise E03

Goal: Make an end-to-end journey with an exception. **Steps:** Draw at least five steps from administrator changing a limit to employee receiving a decision and finance viewing a result. Mark one rejection or review

branch, one handoff, and the distinct outcomes for two users. **Expected:** A journey, not a list of screens. **Hint:** Ask what happens if a rule change is not yet effective.

5. Failure, detection, recovery, trade-off

Failure: Ending the diagram at “request accepted” while finance sees nothing. **Detection:** The last step cannot prove the business outcome. **Recovery:** Walk from first trigger through downstream users and exception resolution. **Trade-off:** A very broad journey exposes hidden work; too much detail obscures the important decisions. Choose one high-value journey, then expand the risky steps.

6. Interview Phrasing

> “I would map the journey from the user’s first action to the business outcome, including approvals, rejected requests, and finance visibility. That reveals handoffs and missing requirements.”

Plain English: The system is only successful when the whole task works. **Follow-up F03:** What if the employee sees ‘accepted’ but finance cannot find the record? **Answer:** AK-E03/F03.

Module B — Identify stakeholders and decision authority (C015)

F01 / L04 / CORE CURRICULUM

L04 — Stakeholder map and affected teams

Checklist: C015 · **Assessment:** A04.

1. Purpose and prerequisites

Purpose: Ask the right people; prevent requirements being approved by people who do not own their impacts. Requires L03 journey and handoffs.

2. Terms and fundamentals

- A **stakeholder** is a person/group with an interest in or impact from a change, even if they never use the software.
- **Decision owner** is the role authorized to accept or reject a defined business decision.
- **Affected team** must change its work or absorb an effect. **Consulted team** provides expertise.
- **Sponsor** backs and funds the initiative. **Subject matter expert (SME)** knows an area well.
- **Accountable** means answerable for the outcome; **responsible** means does the work. **Consulted** gives input; **informed** receives updates. These form **RACI** (Responsible, Accountable, Consulted, Informed) [R1].

3. Worked example — step by step

Fictional stakeholder	Need/concern	Decision or contribution	Affected?
Business sponsor	Lower operating cost	Funds and owns outcome	Yes
Product owner	Usable purchase experience	Prioritizes product needs	Yes
Finance lead	Current, trustworthy records	Defines finance-visibility acceptance	Yes
Risk lead	Review uncertain purchases	Owns risk-policy acceptance	Yes
Security specialist	Restrict access	Advises/approves relevant security controls per governance	Yes
Engineering lead	Deliverable within capacity	Estimates delivery options and technical evidence	Yes

Fictional stakeholder	Need/concern	Decision or contribution	Affected?
Operations/support lead	Diagnosable exceptions	Defines support and escalation needs	Yes
Employees/admins	Complete daily tasks	Validate real journeys	Yes

(1) Extract stakeholders from each journey step. (2) Add teams that govern risk, support, or delivery. (3) Write a need—not just a job title. (4) Distinguish *input* from *approval*. (5) Verify actual decision rights with the organization; table roles are **[Hypothetical practice assumptions]**.

4. Guided exercise E04

Goal: Build an impact map. **Steps:** Name six stakeholder groups for faster transaction visibility. For each, write one need, whether it is affected or consulted, and one question to ask. **Expected:** Include a non-user group. **Hint:** Who handles a record that is displayed incorrectly at 2 a.m.?

5. Failure, detection, recovery, trade-off

Failure: Only the product owner attends discovery. **Detection:** Late objections from risk, finance, support, or users. **Recovery:** Perform impact mapping and scheduled targeted consultation. **Trade-off:** Many reviewers raise quality but can slow decisions; use clear consultation windows and one owner per decision.

6. Interview Phrasing

> "I would map people who use, approve, secure, operate, and depend on the workflow. I would identify who provides input and who has the authority to accept each trade-off."

Plain English: Being affected is not the same as being allowed to approve a risk. **Follow-up F04:** Why should support participate before design? **Answer:** AK-E04/F04.

F01 / L05 / CORE CURRICULUM

L05 — Conflicting objectives and decision rights

Checklist: C015 · **Assessment:** A05.

1. Purpose and prerequisites

Purpose: Reach a justified decision when stakeholder goals collide. Requires L04's stakeholder map.

2. Terms and fundamentals

- **Objective** is a desired result. **Constraint** is a limit that must be respected.
- **Conflict** means two desired outcomes cannot both be fully satisfied under current conditions.
- **Decision right** is explicit authority over a decision. **Escalation** moves unresolved choices to an authorized owner.

- **Risk acceptance** means formally deciding to tolerate a known remaining risk. An architect must not silently accept a business/security risk for its owner.
- **Evidence** means observations, tests, research, or agreed policy that supports a claim.

3. Worked example — step by step

[Hypothetical practice assumptions] Finance wants transaction rows within **30 seconds**; engineering estimates **five minutes** with current capacity; risk requires uncertain items to be marked and reviewed before being called *verified*.

Goal	Why it matters	Potential conflict
Finance: fast visibility	Faster investigations	Data may still be provisional
Engineering: feasible delivery	Cost and reliability	More speed may require more effort
Risk: correct status label	Avoid misleading decisions	Verification can take longer

Reasoning: (1) Ask whether “visible” means *any record* or *verified record*. (2) Keep risk's prohibition on false verified labels as a condition, not a preference to trade away. (3) Compare “fast provisional status” with “slower verified status.” (4) Estimate business consequences and validation steps. (5) Have the designated finance/product decision owner approve the user-experience trade-off, with risk/security approval for their controls. (6) Record unresolved facts and a review trigger.

4. Guided exercise E05

Goal: Produce a decision-rights note. **Steps:** Write two options, each stakeholder's concern, an accountable business decision owner (by role), a risk/control reviewer, one experiment, and a condition for revisiting. **Expected:** No silent decision by the architect. **Hint:** A label such as “pending verification” may separate speed from certainty.

5. Failure, detection, recovery, trade-off

Failure: Engineering declares five minutes “acceptable” without customer approval. **Detection:** No business evidence or acceptance sign-off. **Recovery:** Present choices with measurable impact to the authorized owner. **Trade-off:** Fast provisional display aids finance but requires clear labeling, strong permissions, and later correction; delayed confirmed display simplifies interpretation but harms timely visibility.

6. Interview Phrasing

> “When objectives conflict, I turn each request into a measurable outcome or a hard constraint. I compare feasible options, make residual risk visible, and ask the accountable owner to approve the business trade-off.”

Plain English: Show the real choices and get agreement from the right people. **Follow-up F05:** Can the finance lead override a mandatory security restriction? **Answer:** AK-E05/F05.

Module C — Turn uncertainty into testable requirements (C016)

F01 / L06 / CORE CURRICULUM

L06 — Elicitation: discover before proposing

Checklist: C016 · **Assessment:** A06.

1. Purpose and prerequisites

Purpose: Find real needs rather than accepting the first requested technology. Requires journeys (L03) and stakeholders (L04).

2. Terms and fundamentals

- **Elicitation** means drawing out information from stakeholders, observation, documents, and examples. It is not merely asking for a feature list [R1].
- **Current state** means how work operates today. **Desired state** is the improved future.
- **Open question** has no verified answer yet. **Assumption** is treated as true temporarily but is unverified [R1].
- **Probe** is a follow-up question to uncover details. **Acceptance** means a defined owner agrees a result meets the stated need.

3. Worked example — step by step

A stakeholder says, “We need real-time transaction visibility.” A weak reply is, “We will build streaming dashboards.” That selects a technique without confirming the problem.

Better discovery: (1) “What business action depends on seeing the transaction?” (2) “Which roles need it?” (3) “When does the clock start?” (4) “Does pending information help?” (5) “How old is data today?” (6) “What happens during an exception?” (7) “What is the acceptable delay?” (8) “Which status must be verified?” (9) “Which team is accountable?” (10) “How will we measure benefit?”

Why these matter: Answering (1) reveals value; (2) reveals users; (3)/(7) define a measurable interval; (4)/(8) separate speed from correctness; (5) supplies a baseline; (6) covers failure; (9) secures governance; (10) defines success.

4. Guided exercise E06

Goal: Write *your own* ten clarifying questions for a new complaint: “Managers spend too much time reviewing purchases.” **Steps:** Include current workflow, users, cost/delay baseline, rule exceptions, compliance, priority, outcome, and owner. **Expected:** Ten open questions, not ten technology proposals. **Hint:** Ask “how often” and “what happens if.”

5. Failure, detection, recovery, trade-off

Failure: Asking leading questions ("Do you want a real-time message bus?"). **Detection:** Stakeholders agree to a design they cannot evaluate. **Recovery:** Rephrase around tasks, current pain, acceptable outcomes, and exceptions. **Trade-off:** Workshops expose cross-team conflicts quickly but some users speak less; combine workshops with brief one-to-one interviews and direct observation when possible.

6. Interview Phrasing

> "Before proposing a design, I would clarify the user action, the current delay, the consequence of that delay, and the conditions under which data can be treated as correct."

Plain English: Find what is wrong and what "fixed" means. **Follow-up F06:** What if stakeholders cannot give an exact target? **Answer:** AK-E06/F06.

F01 / L07 / CORE CURRICULUM

L07 — Functional requirements and acceptance criteria

Checklist: C016 · **Assessment:** A07.

1. Purpose and prerequisites

Purpose: State behavior clearly enough to test. Requires L06's discovery notes.

2. Terms and fundamentals

- **Business requirement:** a business goal or result. **Stakeholder requirement:** need of a stakeholder group.
- **Functional requirement:** behavior or information the solution must provide. **Non-functional requirement/quality attribute:** how well it must perform.
- **Acceptance criteria:** observable conditions that must hold for a requirement to be accepted.
- **Actor:** user or external party acting in a task. **Trigger:** event that starts behavior. **Outcome:** observable result.

[Verified external fact] IIBA separates business, stakeholder, and solution requirements (including functional and non-functional), plus transition requirements [R2]. This guide focuses on the types needed for F01.

3. Worked example — step by step

Business objective [hypothetical]: Reduce hours finance spends searching for purchase records.

Stakeholder need: Finance must find a purchase's latest *known* status.

Functional requirement FR-01: "When an authorized finance user searches for a purchase identifier, the system shall display its known status, the time that status was last updated, and whether it is provisional or verified."

Acceptance criteria: (a) An authorized finance user can find a known test purchase; (b) the display contains a status and timestamp; (c) a provisional status is distinguishable from verified; (d) an unauthorized user cannot see the record; (e) unknown identifiers return a defined 'not found' result. **Note:** Authorization is mentioned as a business acceptance concern, not a detailed security design.

Walkthrough: The word *when* gives the trigger; *authorized finance user* gives the actor; *search* gives the behavior; *status/time/label* gives output; criteria make the claim testable. A target such as "within 120 seconds" belongs in a **quality scenario**, not hidden in FR-01.

4. Guided exercise E07

Goal: Write two functional requirements and three criteria for each. **Steps:** Write one for an administrator editing a spending limit and one for an employee receiving a blocked-request explanation. Include actor, trigger, behavior, output, and one misuse case. **Expected:** Requirements do not prescribe a programming language or infrastructure. **Hint:** "When [actor] does [task], the system shall [result]."

5. Failure, detection, recovery, trade-off

Failure: "System shall provide a fast, easy-to-use dashboard." **Detection:** No actor, trigger, measurable test, or clear output. **Recovery:** Split into functional behavior and measurable quality scenarios. **Trade-off:** Over-detailed requirements constrain good design choices; under-detailed requirements permit incompatible interpretations. Specify business-observable behavior and agreed acceptance results.

6. Interview Phrasing

> "I write functional requirements around an actor, a trigger, and an observable result. I add acceptance criteria so the business and delivery team can agree what correct behavior means."

Plain English: Say who needs what and how you will check it. **Follow-up F07:** Is "response in two seconds" a functional requirement? **Answer:** AK-E07/F07.

F01 / L08 / CORE CURRICULUM

L08 — Constraints, assumptions, open questions, and validation

Checklist: C016 · **Assessment:** A08.

1. Purpose and prerequisites

Purpose: Prevent guesses from silently becoming architecture commitments. Requires L07's requirements and L05's decision ownership.

2. Terms and fundamentals

- **Constraint:** a non-negotiable boundary, *if confirmed*, such as a fixed launch date or a policy forbidding unverified results from being labeled verified.
- **Assumption:** unconfirmed working belief. It requires an owner, evidence plan, and a date/review trigger.

- **Open question:** information that must be obtained before making a commitment.
- **Dependency:** another team, activity, or external event needed for success.
- **Risk:** a possible harmful consequence under uncertainty. **Validation:** confirm a requirement contributes to the business value; **verification:** check that it is clearly and correctly stated [R1].

3. Worked example — step by step

ID	Type	Statement	Evidence needed / accountable role
R-01	Requirement	Authorized finance can see a timestamped status	Finance acceptance test / Finance lead
CO-01	Proposed constraint	Unverified status must not be labeled “verified”	Risk policy confirmation / Risk lead
AS-01	Assumption	Most new transaction records can be displayed within 120 seconds	Sample journey data / Engineering + Finance
OQ-01	Open question	Does finance need provisional items or only reviewed records?	Interview / Finance lead
DP-01	Dependency	Source of event timestamps must be identified	Data/system owner
RK-01	Risk	Bad status labels could cause wrong finance action	Review/control test / Risk + Finance

Walkthrough: (1) Tag each statement. (2) Never present proposed constraint CO-01 as an official policy before confirmation. (3) Assign a real role, not a fictitious person. (4) Record when to revisit. (5) Validate requirements against the journey and verify clarity with test cases.

4. Guided exercise E08

Goal: Classify eight statements. **Steps:** Label each as requirement, constraint, assumption, or open question, then name its confirmation owner: (a) “Finance shall see a status timestamp”; (b) “The release date cannot move, per sponsor approval”; (c) “We think all users have stable connectivity”; (d) “Who owns policy approval?”; (e) “Risk checks are always instant”; (f) “Employees shall see a reason when declined”; (g) “Company policy forbids cross-company record access, pending policy review”; (h) “How many purchases arrive at peak?” **Expected:** Two requirements; one confirmed constraint; two assumptions; two open questions; one *proposed/unverified constraint*. **Hint:** Unverified policy claims must be marked as such.

5. Failure, detection, recovery, trade-off

Failure: Treating “risk checks are instant” as an agreed fact. **Detection:** No source or observation. **Recovery:** Log assumption, seek timing evidence, design the decision to tolerate uncertainty. **Trade-off:** Early assumptions let teams explore options; unchecked assumptions become expensive rework. Use short decision gates instead of stopping all planning.

6. Interview Phrasing

> "I keep requirements, hard constraints, assumptions, and unanswered questions in separate lists. Each important assumption has an owner and a way to validate it before we commit to a design."

Plain English: A guess is not a fact. **Follow-up F08:** What if the launch date and security policy conflict? **Answer:** AK-E08/F08.

Module D — Write measurable quality attributes (C017)

F01 / L09 / CORE CURRICULUM

L09 — Quality attributes and scenario structure

Checklist: C017 · **Assessment:** A09.

1. Purpose and prerequisites

Purpose: Replace “fast,” “always available,” “secure,” and “easy to change” with scenarios that can guide choices and be tested. Requires L07 requirements and L08 uncertainty labels.

2. Terms and fundamentals

- A **quality attribute** describes how well a system serves a purpose, not merely one feature. Examples are availability, performance, security, freshness, and changeability.
- A **quality-attribute scenario** states what happens, under which conditions, and how success is measured.
- **Stimulus** is something that happens. **Environment** is the operating condition. **Response** is how the system behaves. **Response measure** is the numeric or observable pass rule.
- **Source** generates the stimulus; **artifact** is what is affected. A common architecture scenario template uses **six fields**: source, stimulus, environment, artifact, response, response measure. We use it as a learning template; we do not claim ISO prescribes this wording.
- **Threshold** is a boundary value. **Measurement window** is the period over which results are counted.

[Verified external fact] Carnegie Mellon Software Engineering Institute (SEI) uses stakeholder-generated scenarios to identify and prioritize architecturally important quality needs before architecture exists [R3]. ISO/IEC 25010:2023 provides a product quality model to specify and evaluate quality [R4]. The complete ISO standard text was not accessed.

3. Worked example — step by step

Vague: “The transaction view must be real time.”

Precise [Hypothetical practice assumption]:

Field	Example
Source	A newly recorded purchase
Stimulus	Its record becomes available at the agreed source boundary
Environment	Normal operation during expected business traffic
Artifact	Finance transaction-status view

Field	Example
Response	The view displays a **provisional** record and its source timestamp
Response measure	At least 99% of eligible records become visible within 120 seconds, measured monthly

Walkthrough: (1) Identify the business need. (2) Define the clock's start and stop. (3) Specify eligible records, including how to count missing records. (4) State operating conditions. (5) Define the response. (6) Include threshold, proportion, period, and measure source. (7) Seek owner agreement before calling it accepted.

4. Guided exercise E09

Goal: Turn “the system should be reliable” into a six-field scenario. **Steps:** Choose a purchase-request decision; define the stimulus, environment, response, and measure. **Expected:** All six fields and a clear observation window. **Hint:** Is “no server crash” the same as successful purchase processing?

5. Failure, detection, recovery, trade-off

Failure: “99% real time” without definitions of events or timing. **Detection:** Two teams compute different results. **Recovery:** Define the six fields and reproducible calculation. **Trade-off:** Stricter targets may improve experience but increase cost; choose targets by evidence and impact.

6. Interview Phrasing

> “I turn quality expectations into scenarios with a trigger, operating condition, response, and measurable result. That lets us compare options against user impact instead of debating adjectives.”

Plain English: Describe exactly what happens and what counts as good enough. **Follow-up F09:** Why specify the clock start and population? **Answer:** AK-E09/F09.

F01 / L10 / CORE CURRICULUM

L10 — Availability: can the user complete the task?

Checklist: C017, availability · **Assessment:** A10.

1. Purpose and prerequisites

Purpose: State when a required journey is usable. Requires L03 journey and L09 scenario template.

2. Terms and fundamentals

- **Availability** is ability to use a service when needed, under an agreed definition.

- A **successful user journey** means the required business action completes correctly, not just that a server responds.
- A **service level indicator (SLI)** measures service behavior. A **service level objective (SLO)** gives a target for that indicator.
- **Denominator** is all eligible attempts. **Numerator** is the attempts satisfying the rule. A **maintenance window** is planned downtime; do not exclude it silently.

[Verified external fact] Google SRE discusses availability SLIs and SLOs tied to critical user functionality [R5]. Detailed SLO operations belong to T07.

3. Worked example — step by step

[Hypothetical practice assumption] At expected load, at least **99.5% of eligible valid employee purchase submissions return a correct, usable business response per calendar month**. The denominator counts valid authorized submissions at the agreed observation boundary. An intentional policy decline **may count as success** if the rule was applied correctly and a usable explanation was shown.

```
Employee submits a request
       |
       v
Correct business result?
    yes = successful journey
    no  = failed journey
```

Example [invented]: 9,950 correct results / 10,000 eligible attempts = 99.5%. Beware of failures that never reach server logs; agree how to observe those too.

4. Guided exercise E10

Goal: Specify availability for editing a policy. **Steps:** Fill all six scenario fields, define eligible attempts and success, and propose a hypothetical target and period. **Expected:** Business-facing pass/fail rule. **Hint:** Returning the wrong saved limit is not success.

5. Failure, detection, recovery, trade-off

Failure: An error screen counts as “available.” **Detection:** User complaints disagree with service health. **Recovery:** Measure the completed journey. **Trade-off:** Higher targets can cost more; never bypass required risk checks to improve the availability score.

6. Interview Phrasing

> “I define availability by whether customers can correctly complete a critical journey. Then I agree on eligible attempts, the observation period, and a measurable target.”

Plain English: A server can respond and the task can still fail. **Follow-up F10:** Is a legitimate policy decline an outage? **Answer:** AK-E10/F10.

F01 / L11 / CORE CURRICULUM

L11 — Latency: how long the user waits

Checklist: C017, latency · **Assessment:** A11.

1. Purpose and prerequisites

Purpose: Express waiting time as a user-facing requirement. Requires L09 and L10's eligible population.

2. Terms and fundamentals

- **Latency** is elapsed time between named start and stop events.
- **End-to-end latency** covers the user experience; **internal processing time** covers one system step.
- A **percentile** indicates a portion of observed timings. **95th percentile (p95)** is a threshold approximately 95% of measured events meet, under a declared calculation method.
- A **timeout** means a waiting limit expired; it does **not** prove downstream processing failed.

[Verified external fact] Google SRE discusses latency percentiles and measurement locations close to user experience [R5].

3. Worked example — step by step

[Hypothetical practice assumption] During expected peak traffic, **95% of valid purchase submissions show a correct initial decision within 2 seconds** of pressing Submit, measured over a rolling 28-day window. A "pending review" response can meet an *initial-response* target but must **not** be counted as a completed approval.

Invented timings: 1.0, 1.2, 1.5, 1.8, 9.0 seconds. Average is 2.9 seconds; only four of five (80%) meet 2 seconds. Five observations are too few to estimate a production p95 reliably.

4. Guided exercise E11

Goal: Replace "fast search" with a latency scenario. **Steps:** Define clock boundaries for finance search; select a hypothetical percentile, threshold, load and window; describe slow requests. **Expected:** Measurable user-level latency. **Hint:** Displaying a spinner is not displaying usable results.

5. Failure, detection, recovery, trade-off

Failure: Timing only backend processing. **Detection:** Users experience much longer screen waits. **Recovery:** Measure full user response as well as internal timings. **Trade-off:** A quick preliminary response can improve responsiveness, but it cannot be called final.

6. Interview Phrasing

> "I measure latency for the action the user actually experiences, use percentiles rather than only averages, and distinguish an early acknowledgement from a final business decision."

Plain English: Time the task, not one hidden part. **Follow-up F11:** Why can an average hide poor experience? **Answer:** AK-E11/F11.

F01 / L12 / CORE CURRICULUM

L12 — Security: who may see or change what?

Checklist: C017, security · **Assessment:** A12.

1. Purpose and prerequisites

Purpose: Express a business-facing security outcome, before technology design. Requires roles L01, owners L04, scenarios L09.

2. Terms and fundamentals

- **Authentication** checks identity. **Authorization** checks permission to perform a specific action.
- **Confidentiality** restricts information to authorized parties. **Integrity** protects from unauthorized modification. **Availability** ensures timely authorized access.
- **Least privilege** means giving only the access needed. **Audit evidence** is a trustworthy record that assists an investigation.
- **Negative test** attempts a forbidden action. Positive tests check permitted actions still work. Neither proves absence of every vulnerability.

[Verified external fact] NIST frames information security with confidentiality, integrity, and availability [R6]. Detailed controls belong in T06.

3. Worked example — step by step

Scenario [Hypothetical practice assumption]: A finance user from organization A attempts to view organization B's record. In normal operation, the transaction view **denies access and returns no details**. All defined automated cross-organization negative tests must pass, with reviewable evidence. This is an **acceptance test**, not a promise of zero production vulnerability.

Walkthrough: (1) Identify the sensitive data. (2) Identify role and forbidden action. (3) State safe response. (4) Define a test set and owner. (5) Add a positive authorized case. (6) Record remaining risk for specialist review.

4. Guided exercise E12

Goal: Write two access scenarios. **Steps:** One permits an administrator to update its organization's limit; one rejects an employee trying to do so. For each specify response, evidence, reviewer. **Expected:** Positive and negative acceptance cases. **Hint:** Being logged in does not imply edit rights.

5. Failure, detection, recovery, trade-off

Failure: Treating login as enough to read every organization's data. **Detection:** Unauthorized cross-company reads in negative tests. **Recovery:** Define per-record authorization requirements and security

review. **Trade-off:** Strong controls may increase complexity or latency; mandatory protection cannot be silently waived for speed.

6. Interview Phrasing

> "I describe security as allowed and forbidden business actions, including cross-company access. I define negative tests and involve the authorized security owner in accepting remaining risk."

Plain English: Say what must never be exposed and test it. **Follow-up F12:** Is login alone enough? **Answer:** AK-E12/F12.

F01 / L13 / CORE CURRICULUM

L13 — Data freshness: how recent is the displayed fact?

Checklist: C017, data freshness · **Assessment:** A13.

1. Purpose and prerequisites

Purpose: Make "real time" precise for finance. Requires journey L03 and scenario L09.

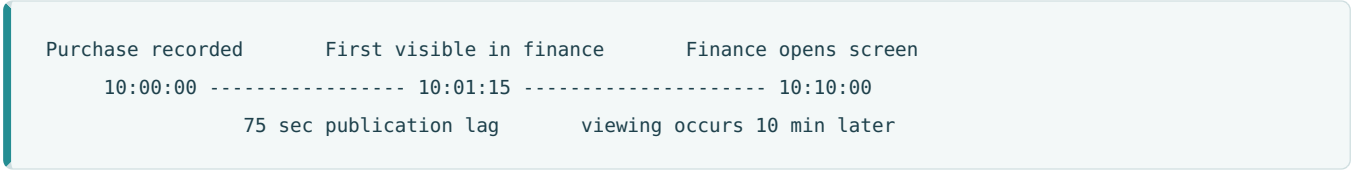
2. Terms and fundamentals

- **Event time** is when an event occurs at a defined source. **Source timestamp** is the recorded event time.
- **First-visible time** is when a user can first see the relevant record.
- **Publication lag** = first-visible time − source time. **Displayed age** = viewing time − timestamp of the latest relevant update. These are **different** measures.
- **Freshness** means how current the displayed data is. **Completeness** means whether expected records are present; a fresh report can omit important items.
- **Provisional** information is not fully verified. **Stale** data exceeds the accepted freshness threshold.

[Verified external fact] Google SRE discusses freshness and correctness measures for data pipelines [R5]. The data model and reconciliation procedures belong to T05/D02.

3. Worked example — step by step

Scenario [Hypothetical practice assumption]: A purchase becomes eligible for reporting at an agreed source boundary. Under normal load, the finance view shows a correctly labeled provisional or verified row **within 120 seconds for at least 99% of eligible records each month**. The view shows a timestamp and a freshness warning when the threshold is breached.



Walkthrough: (1) Define the source event. (2) Define first visibility. (3) Count *all eligible records*, including those not visible. (4) Mark provisional versus verified. (5) Compute lag; note viewing-time age is separate. (6) Ask whether missing/incorrect rows should block acceptance.

4. Guided exercise E13

Goal: Distinguish two freshness measures. **Steps:** A record occurs at 10:00, first appears at 10:04, and is viewed at 10:12 without any later update. Calculate publication lag and displayed age. Explain how to count a record that never appears, and give one quality safeguard. **Expected:** Two different elapsed times and treatment of missing records. **Hint:** Missing items need their own failure count.

5. Failure, detection, recovery, trade-off

Failure: Measuring only records already displayed. **Detection:** Missing items disappear from the denominator. **Recovery:** Compare known eligible event identifiers with visible records; report delayed *and missing* counts. **Trade-off:** Quick provisional visibility helps finance, but status labels, subsequent correction, and access controls must prevent misleading use.

6. Interview Phrasing

> "I ask which event starts the clock and how current the finance view must be. I measure missing records as well as delays, and distinguish provisional status from verified status."

Plain English: A quick report still fails if it silently misses purchases. **Follow-up F13:** Can data be fresh but incorrect? **Answer:** AK-E13/F13.

F01 / L14 / CORE CURRICULUM

L14 — Changeability: can business rules evolve safely?

Checklist: C017, changeability · **Assessment:** A14.

1. Purpose and prerequisites

Purpose: Express "easy to change" as a measurable business requirement. Requires capabilities L02, requirements L07, scenarios L09.

2. Terms and fundamentals

- **Changeability** here is the ease and safety of changing needed business behavior. **Modifiability** is a related architecture term [R3]; they are not asserted to be identical formal standard labels.
- **Lead time** is elapsed time between explicitly defined change start/end points.
- **Change scope** says which rules or outputs are affected. **Regression** is an unintended break in previously correct behavior.
- **Rollback** reverses a problematic change. **Configuration** stores behavior-affecting values separately from code; it is one possible implementation, not an F01 mandate.

3. Worked example — step by step

Scenario [Hypothetical practice assumption]: After an authorized administrator's standard spending-limit update has completed required approval, the new validated limit becomes effective **within one business day for 95% of eligible changes per quarter**, with no change to unrelated policies. Previous and new values remain inspectable for authorized review.

Walkthrough: (1) Define the change class: *standard limit value*, not arbitrary new rules. (2) Start the clock after required approvals. (3) End when business behavior actually uses the new value. (4) Verify unrelated rules still work. (5) Decide who may reverse an unsafe change. (6) Confirm that the business values this time target.

4. Guided exercise E14

Goal: Write a changeability scenario for adding an optional approval step. **Steps:** Define “small change,” clock start/end, proposed target, a regression check, and rollback trigger. **Expected:** A concrete change with measurable outcome and safety checks. **Hint:** “Code deployed” is not the same as “new rule works correctly.”

5. Failure, detection, recovery, trade-off

Failure: Promising “all changes in one day” without considering safety approvals and scope. **Detection:** Complex changes routinely fail targets or bypass checks. **Recovery:** Define separate change classes and necessary reviews. **Trade-off:** Configuration can shorten standard changes but increases the number of rule combinations that must be controlled and tested.

6. Interview Phrasing

> “I define changeability using a specific class of business change, its approval point, the time until it becomes effective, and how we avoid regression. I would not give one estimate for every change.”

Plain English: Ask how quickly a *specific* change can happen safely. **Follow-up F14:** When can configurability increase risk? **Answer:** AK-E14/F14.

F01 / L15 / CORE CURRICULUM

L15 — Prioritize quality by business impact

Checklist: C017, ranking · **Assessment:** A15.

1. Purpose and prerequisites

Purpose: Decide what is truly important when not every target can be maximized. Requires L10–L14's five scenarios and L05's decision rights.

2. Terms and fundamentals

- **Business impact** is the effect on customers, operations, financial results, or obligations if a quality target is missed.
- **Severity** is how harmful a failure would be. **Likelihood** describes how likely it is. **Urgency** means how quickly action is needed.
- **Hard constraint** is not freely traded. **Priority** guides investment among negotiable improvements.
- **Impact ranking** is a defensible ordering, *not* a substitute for testing or approval.
- **Revisit trigger** is a change in business or technical facts that causes reassessment.

[Verified external fact] SEI's Quality Attribute Workshop prioritizes stakeholder quality scenarios using business drivers [R3].

3. Worked example — step by step

[Hypothetical practice assumptions; this is one defensible ranking, not a company policy]:

Rank	Quality goal	Business failure if unmet	Status / rationale
Constraint	Security	Cross-company exposure or false verified label	Must pass agreed security/ control review; not traded away
1	Data freshness	Finance cannot investigate purchases in time	Directly addresses stated complaint
2	Availability	Employees or finance cannot complete critical tasks	High business disruption
3	Latency	Employees wait too long for decisions	Important, but early pending responses may help
4	Changeability	Standard rule updates take too long	Valuable, possibly less urgent for initial fix

Walkthrough: (1) Identify the business complaint and harm. (2) Separate mandatory controls from negotiable target levels. (3) Ask each stakeholder for examples of impact. (4) Compare feasibility and costs of improvements. (5) Get decision-owner confirmation. (6) Revisit if major new evidence emerges. A different business case may rank changeability first.

4. Guided exercise E15

Goal: Rank the five qualities under a changed scenario. **Steps:** Suppose [Hypothetical practice assumption] a client updates policies daily, but finance reviews records only once a week. Rank the top three, state business evidence you need, and flag at least one non-negotiable security constraint. **Expected:** Reasoned order with uncertainty and owner. **Hint:** A requirement's business urgency changes with the workflow.

5. Failure, detection, recovery, trade-off

Failure: The architect assigns priorities alone using invented weights. **Detection:** No customer impact, stakeholder validation, or measurement. **Recovery:** Compare consequence, affected population, feasible

targets, and cost, then get agreement. **Trade-off:** Improving freshness may raise complexity or show provisional data; improving certainty may delay visibility. Ranking selects an investment focus, not permission to break controls.

6. Interview Phrasing

> "I prioritize quality attributes by the cost of failing the business journey, not by technical fashion. I treat required security controls as constraints, compare the negotiable targets, and confirm the ranking with the accountable owner."

Plain English: Rank the pain and risk, not favorite technologies. **Follow-up F15:** Can security be below freshness in the ranking? **Answer:** AK-E15/F15.

Module E — Prove business and technical success (C018)

F01 / L16 / CORE CURRICULUM

L16 — Business KPIs, baselines, and ownership

Checklist: C018 · **Assessment:** A16.

1. Purpose and prerequisites

Purpose: Decide if the solution helped the business, not just whether software was delivered. Requires L01 users, L03 journeys, and L07 requirements.

2. Terms and fundamentals

- A **key performance indicator (KPI)** is a selected measure of progress toward an important business outcome.
- **Metric** is any measure. A metric only becomes a meaningful KPI when tied to a goal and owner.
- **Baseline** is an agreed measurement of current results before a change.
- **Target** is the desired future value, with a deadline and measurement rule.
- **Leading measure** can signal future results (e.g., faster status availability). **Lagging measure** shows an outcome after it occurs (e.g., fewer finance hours spent investigating).
- **Countermetric/guardrail** checks whether improving one result harms another (e.g., faster visibility but more incorrect status).
- **Unit**, **population**, **measurement window**, **data source**, and **owner** make a KPI usable.

[Verified external fact] IIBA defines metrics as quantifiable indicators and separates business objectives from solution requirements [R1, R2].

3. Worked example — step by step

Business goal [hypothetical]: Reduce manual transaction investigations for Northstar finance.

Measure	Type	Hypothetical example definition	Proposed owner
Weekly minutes spent locating transactions	Business KPI (lagging)	Minutes per finance analyst-week, measured via sampled time logs	Finance lead
Percentage of transactions located without support	Business KPI	Successful independent lookups / eligible lookup attempts	Finance lead

Measure	Type	Hypothetical example definition	Proposed owner
Record-visible-within-120s ratio	Technical freshness measure	Eligible records visible within 120s / all eligible source records	Engineering + Finance
Wrong status-label rate	Guardrail	Confirmed incorrect labels / reviewed records	Risk + Finance
Access violation test results	Security guardrail	Unauthorized test cases exposing any data	Security lead

[Hypothetical practice assumption] Baseline is **90 minutes per analyst-week**; proposed target **45 minutes within 90 days** of launch. These are **invented**, not measured. Do not claim success until a real baseline and follow-up measurement exist.

Walkthrough: (1) Name the business outcome. (2) Choose a measure affected by that outcome. (3) Define what counts in numerator and denominator or measurement unit. (4) Obtain a baseline. (5) Agree target and time period. (6) Name accountable owner. (7) Add guardrails to detect harmful shortcuts.

4. Guided exercise E16

Goal: Write a KPI card. **Steps:** Pick the outcome “less time investigating missing transactions.” Provide business measure, exact calculation or time-log rule, baseline collection plan, hypothetical target, time period, owner, and guardrail. **Expected:** Can be computed without guessing. **Hint:** Number of dashboards opened is not the outcome.

5. Failure, detection, recovery, trade-off

Failure: Calling “number of deployed features” a business KPI. **Detection:** It can rise while finance effort stays high. **Recovery:** Tie the KPI to the user's job and business outcome. **Trade-off:** A simple KPI is easier to maintain but may miss nuance; combine a few meaningful outcome and guardrail measures, not dozens of vanity metrics.

6. Interview Phrasing

> “I would agree on a business KPI before choosing the technology, capture the baseline, set a time-bound target, and pair it with technical indicators and safety guardrails.”

Plain English: Decide what business improvement must be proven, and check that shortcuts do not cause harm. **Follow-up F16:** Why is deployment count a poor measure of value? **Answer:** AK-E16/F16.

F01 / L17 / CORE CURRICULUM

L17 — Measurement design: link technical signals to business outcomes

Checklist: C018 · **Assessment:** A17.

1. Purpose and prerequisites

Purpose: Show what to measure, how to compute it, and what evidence supports a claim of success. Requires L13 freshness and L16 KPIs. **Python is optional**; the code is a tiny learning tool, not an F01 programming requirement.

2. Terms and fundamentals

- An **event record** stores a fact and time. A **timestamp** is a recorded date/time; a **time zone** prevents ambiguous clock values.
- A **data collection plan** states how records are sampled and checked.
- A **leading indicator** (freshness) can improve before the **business outcome** (reduced finance effort). Correlation alone does not prove causation.
- **Segmentation** means checking groups separately (customer type, team, high-volume periods) so averages do not hide poor experience.
- **Validation evidence** may include direct observation, controlled pilots, matched before/after samples, and negative-case tests.

[Verified external fact] Google SRE distinguishes end-to-end user-oriented signals from narrower server measures and describes pipeline freshness indicators [R5, R8].

3. Worked example — metric definition, code, and line-by-line explanation

[Hypothetical practice assumption] This dataset contains three eligible purchase records. We will check how many first became visible within **120 seconds**. For missing records in a production measure, they must remain in the source population and count as failures; this miniature example has none missing.

Save the following as `freshness_example.py`, then run `python freshness_example.py` (a Python installation is needed; the code uses the standard library only).

```
from datetime import datetime

records = [
    ("P1", "2026-10-01T10:00:00+00:00", "2026-10-01T10:01:00+00:00"),
    ("P2", "2026-10-01T11:00:00+00:00", "2026-10-01T11:03:30+00:00"),
    ("P3", "2026-10-01T12:00:00+00:00", "2026-10-01T12:01:30+00:00"),
]

def publication_lag_seconds(event_time, visible_time):
    start = datetime.fromisoformat(event_time)
    end = datetime.fromisoformat(visible_time)
    if start.tzinfo is None or end.tzinfo is None:
        raise ValueError("Timezone required")
    lag = (end - start).total_seconds()
    if lag < 0:
        raise ValueError("Visible time is before event time")
    return lag
```

```
lags = [publication_lag_seconds(event, visible) for _, event, visible in records]
on_time = sum(lag <= 120 for lag in lags)
rate = on_time / len(records)
print("Publication lag (seconds):", lags)
print("Within 120 seconds:", f"{rate:.1%}")
assert lags == [60.0, 210.0, 90.0]
assert rate == 2 / 3
```

Explain every executable line or block:

1. `from datetime import datetime` imports the standard-library timestamp parser.
2. `records = [` starts a small collection; each row contains ID, event time, first-visible time.
3. The `P1` row is 60 seconds late; `+00:00` means its time is explicitly in the UTC offset zero time zone.
4. The `P2` row is 210 seconds late.
5. The `P3` row is 90 seconds late.
6. `]` closes the list.
7. `def publication_lag_seconds(...)` defines a reusable calculation from two timestamps.
8. `start = ...` parses the source timestamp.
9. `end = ...` parses the first-visible timestamp.
10. `if ... tzinfo ...` checks that both timestamps contain time-zone information.
11. `raise ValueError(...)` rejects ambiguous timestamps if the check fails.
12. `lag = ... total_seconds()` subtracts start from end and converts the duration to seconds.
13. `if lag < 0` finds an impossible negative lag in this simplified example.
14. `raise ValueError(...)` stops on this data-quality problem instead of treating it as fast performance.
15. `return lag` gives the validated number back to the caller.
16. `lags = [...]` processes each record once. The underscore `_` discards the identifier because this example only needs times.
17. `on_time = sum(...)` counts records with lag no larger than 120 seconds; `True` behaves as 1 here.
18. `rate = ...` divides successes by eligible rows. A production implementation must also handle an empty population and missing records.
19. First `print(...)` displays raw lag values so they can be inspected.
20. Second `print(...)` displays the success ratio as a percentage to one decimal place.
21. First `assert` verifies the exact expected lag list; if false, execution stops with an assertion error.
22. Second `assert` checks the fraction is two thirds.

Expected output:

```
Publication lag (seconds): [60.0, 210.0, 90.0]
Within 120 seconds: 66.7%
```

Business interpretation: Only two of three fictional events meet the sample 120-second threshold. That alone **does not** show the finance team's hours fell. The business KPI needs its own baseline and outcome study. **Important limitation:** A genuine production freshness SLI must be based on an authoritative eligible-

event inventory and an explicit policy for missing, duplicate, invalid, and late timestamps, plus segmentation and data-quality checks. This code is not production monitoring.

4. Guided exercise E17

Goal: Evaluate a changed threshold and a missing row. **Steps:** (a) Without code, compute how many of these three records meet 240 seconds; (b) propose a fourth eligible record with no visible timestamp and define how it affects the denominator; (c) identify a separate finance business KPI and an adverse-effect guardrail; (d) optionally modify the code to compute only part (a). **Expected:** Honest calculation and data-quality caveat. **Hint:** A missing row should never improve the success percentage.

5. Failure, detection, recovery, trade-off

Failure: "Freshness improved, therefore finance productivity improved." **Detection:** No before/after finance outcome evidence; sample sizes or case mix changed. **Recovery:** Collect comparable finance effort measurements, check status accuracy, segment users, and inspect explanations other than the software change. **Trade-off:** More detailed measurement improves confidence but increases data-collection cost and privacy exposure; measure only what is necessary and obtain appropriate approval.

6. Interview Phrasing

> "I would instrument the user journey and connect freshness and correctness measures to a business outcome such as finance investigation time. I would establish a baseline, use guardrails, and validate whether the improvement actually occurred."

Plain English: Better technology measurements are useful evidence, but not proof of business value. **Follow-up F17:** What if freshness improves while finance time gets worse? **Answer:** AK-E17/F17.

Module F — Independent application and senior-level judgment

The F01 one-page problem brief — cumulative exercise CE01

[Source requirement] For the running case, produce a one-page problem brief, ask **ten clarifying questions**, then rank the **top three quality attributes** with business reasons. This is the required F01 card practice. **Do not read the model solution until you have produced your own.**

Required one-page structure

Heading	What to write
Problem and business goal	What is wrong today? What should change for the business?
Customer types / users	Name at least two customer contexts and three operating roles
End-to-end journey	Trigger, at least five steps, successful outcome, one exception
Capabilities	At least four business abilities; no software-component choices
Stakeholders and decision rights	Sponsor, finance, risk/security, product, engineering, operations, user validation
Requirements	Three testable functional requirements and their acceptance criteria
Uncertainty log	At least two confirmed/proposed constraints, three assumptions, three open questions; clearly label verification status
Quality attribute scenarios	One each for availability, latency, security, data freshness, and changeability; include conditions and measures
Top-three quality priorities	Business impact, constraint treatment, conflict/trade-off, approval owner
Success measures	Two business KPIs, two technical measures, at least one guardrail, baseline/target/window/owner
Evidence and next decision	Which facts need verifying, which experiment, who decides, when to revisit

Deliverable CE01: One page for the executive-facing brief **plus** an appendix containing the full scenarios, journey, questions, and measurement definitions. Do not shrink required evidence merely to fit one page. Place a one-paragraph recommendation at the top, but no component design.

Ten clarifying-question prompts (answer by writing your own questions)

1. Business consequence of a late transaction?
2. Distinct needs of an employee and finance user?
3. Current source of delays and baseline?
4. Difference between first visible and verified?
5. Start and end of each proposed clock?
6. Missing or incorrect records?
7. Access boundary between customer organizations?
8. What can the business tolerate when risk checks are pending?
9. Who owns approval and escalation?
10. Which business KPI proves the outcome?

Evaluation CE01: Must include **10 fully written questions**, not just these prompts; five testable quality scenarios; an owned uncertainty log; rank top three with explicit business reasons; and a defensible measurement approach. Compare with **AK-CE01** only after drafting.

Practice questions PQ01–PQ20

Answer aloud or in writing **without notes**. Mark the primary checklist item in brackets.

ID	Question	Item
PQ01	A company buys a service, but staff operate it. Who is customer and who is user?	C014
PQ02	Differentiate capability, process, and feature using a spend-control example.	C014
PQ03	Trace a purchase journey from request through a finance exception. Where are handoffs?	C014
PQ04	Name a non-user stakeholder for transaction visibility and why they matter.	C015
PQ05	Finance, engineering, and risk disagree. Who contributes and who approves?	C015
PQ06	What is wrong with accepting “we need streaming” as the first requirement?	C016
PQ07	Write one functional requirement and two acceptance tests for finance search.	C016
PQ08	Classify “risk checks take zero time” when nobody has confirmed it.	C016

ID	Question	Item
PQ09	Explain verification versus validation of a requirement.	C016
PQ10	Give all six fields of a measurable quality-attribute scenario.	C017
PQ11	Is a correct business decline a service outage? Explain your denominator.	C017
PQ12	Why measure user-facing p95 instead of only mean backend latency?	C017
PQ13	State a positive and negative authorization scenario.	C017
PQ14	Explain publication lag versus displayed age, with clocks.	C017
PQ15	Write a changeability target for routine policy updates, without skipping approval.	C017
PQ16	Rank three quality attributes for a time-critical finance review and defend your order.	C017
PQ17	Business KPI versus technical measure: give one linked pair.	C018
PQ18	What must be defined before claiming "99% of records visible in two minutes"?	C018
PQ19	Why can a faster dashboard fail to improve business results?	C018
PQ20	A 5-minute option is cheaper, a 30-second option is faster, but risk has concerns. How do you decide?	All

PQ scoring: For each question award 0 = no accurate answer, 1 = correct explanation, 2 = explanation with a concrete example, 3 = explanation + example + justified trade-off/evidence. Maximum **60**. This diagnostic practice score does **not** replace the standard 1–5 card rubric or the independent senior decision test. Answer sketches: **AK-PQ**.

Mandatory F01 senior decision challenge — SD01

[Source requirement] Original F01 decision test: **Finance says "real time," engineering proposes a five-minute delay, and risk wants stronger checks. Define measurable alternatives and who approves the business trade-off.**

Case facts: Only the three stated positions are known. **[Hypothetical practice assumptions, to be tested rather than claimed]:** Finance sometimes acts on provisional transaction status; risk may delay verification;

engineering can compare a 30-second provisional target and a five-minute verified target. The purchase record is **not** evidence of final financial settlement. Do **not** infer American Express technology, contractual targets, or policy.

Your response must contain all seven required elements

1. **Business goal:** Who needs what business outcome? State “visible” versus “verified.”
2. **Known/assumed/open:** Separate observed facts, provisional numerical assumptions, and missing evidence.
3. **At least two feasible options:** Include measurable percentages, windows, status meanings, and failure behavior.
4. **Recommendation and trade-off:** Compare speed, correctness, risk, feasibility, costs, and user consequences.
5. **Failure, security, and data implications:** Missing rows, misleading provisional labels, wrong access, delayed risk decisions, record-quality controls.
6. **Evidence/experiment:** Interview users; baseline; realistic pilot/sample; compare late/missing/wrong-status rates and real finance effort.
7. **Owners and measurable outcomes:** Name a business approver, control approver, engineering evidence owner, support/data owner, business KPI, technical measures, decision revisit point.

Decision template (fill before opening the answer key)

Field	Your answer
Business goal and user	
Known facts / practice assumptions / open questions	
Option A (targets, labels, failure cases)	
Option B (targets, labels, failure cases)	
Optional Option C / phased approach	
Decision and main sacrifice	
Mandatory security/risk guardrails	
What to measure / evidence needed	
Accountable business approver	
Risk/security control approver	
Technical/data/operations owners	
Review trigger	

Interview simulation: Give a two-minute recommendation. Then respond to each change **without changing the meaning of “verified”**: (i) risk checks now take 15 minutes for 5% of requests; (ii) finance reports

missing transactions after deployment; (iii) the sponsor cuts delivery capacity by 30%. The numbers are **fictional change prompts**, not observed facts. This is a standalone **untimed practice challenge**, not X01/X02 timed-mock mode.

Pass standard: Independently achieve **at least 4/5 on every applicable standard dimension** in the rubric near the end. The answer key contains one possible model defense; reasonable alternatives may also pass if evidenced.

Answer key — open only after practicing

The examples below are **model exercises, not actual evidence** of Northstar or an employer. Multiple reasoned answers can be correct. Each gives a model result, a correction point, and the interview follow-up answer.

AK-P0 — prerequisite check

1. **Goal**: reduce finance effort; **feature**: a search screen. The screen can exist and the effort might not fall.
2. Employee wants quick next action; finance wants trusted transaction status. Both can use one service differently.
3. "Fast" lacks a named action, time boundary, threshold, population, and measurement window.

AK-E01/F01 — customer types and roles

Model: Customer: a university administration buying purchasing controls. User role 1: department manager requests purchases and wants a clear response. User role 2: central finance reviewer checks spending and wants reliable records. Question: "May the department manager approve their own request?" **Typical correction:** Do not assume that a role name grants permissions. **Follow-up:** Yes, one person may act in multiple roles, but applicable permissions and separation-of-duties rules require validation.

AK-E02/F02 — capabilities

Model: Manage spending policy (administrator defines limit); check requests (employee gets decision); display purchase status (finance reviews status); manage exceptions (support/finance investigates a missing record). **Typical correction:** "Build a React screen" is not a business capability. **Follow-up:** Yes, several systems may support one business capability; F01 does not decide their boundaries.

AK-E03/F03 — journey

Model: Admin proposes limit → authorized approval → new value becomes effective → employee submits → rule evaluated → allowed / blocked / held for review → outcome shown → finance receives status → missing status sent for investigation. **Roles:** Admin, employee, reviewer, finance. **Handoff:** Review decision or finance status. **Typical correction:** A request receipt does not prove completion. **Follow-up:** Finance visibility failure is an unresolved journey; measure it and provide an owned exception route.

AK-E04/F04 — stakeholders

Model: Finance lead (status quality; affected); employees (feedback; affected); administrators (policy change; affected); risk lead (correct verification label; affected); security team (data access; consulted/ control owner); engineering lead (feasibility; affected); support lead (recovery; affected). Ask each about

failure impact and acceptance. **Typical correction:** Excluding operations hides 24-hour support needs. **Follow-up:** Support knows how broken or disputed records reach human repair, and whether the design is diagnosable.

AK-E05/F05 — conflict and owners

Model: A = five-minute verified-only view, simpler status interpretation but slow. B = a provisional row within a proposed 60 seconds, then update after completed risk checks; faster but additional correctness/access controls. Product/finance sponsor approves business user-experience and cost trade-off; authorized risk/security owners must approve applicable controls; engineering supplies estimates/tests; support owns exception procedures. Pilot both with fake data. **Typical correction:** Engineering cannot accept all business/security risks alone. **Follow-up:** No. Mandatory policies or legal obligations cannot be waived by an unrelated business owner; follow formal control governance.

AK-E06/F06 — elicitation

Model ten questions: (1) What review work takes the most time? (2) Which teams review it? (3) How many cases per week? (4) What is the current median and long-tail review time? (5) Which policies trigger review? (6) What information is missing? (7) How often is an exception legitimate? (8) What happens when review is delayed? (9) Who can change or approve rules? (10) What reduction in effort would count as success? **Typical correction:** Avoid "Would a new dashboard solve it?" **Follow-up:** Propose candidate targets as assumptions; collect baseline or run a low-cost user study before approval.

AK-E07/F07 — functional requirements

FR-A model: "When an authorized administrator submits an approved new limit, the system records it and shows its effective value/time." Tests: correct value after approval; unauthorized edit is rejected; invalid/negative limit produces defined rejection. **FR-B:** "When an employee request is blocked by a spending rule, the system shows a permitted reason and a next action." Tests: matched reason; no exposure of restricted policy details; request remains blocked. **Typical correction:** Do not force a database/technology choice into the requirement. **Follow-up:** "Response within two seconds" is a measurable quality/latency requirement; the behavior itself is functional.

AK-E08/F08 — uncertainty classification

Model: (a) requirement; (b) confirmed constraint **only if sponsor approval is real and recorded**; (c) assumption; (d) open question; (e) assumption; (f) requirement; (g) proposed/unverified constraint because policy review is pending; (h) open question. **Typical correction:** Claimed approvals without evidence remain unverified. **Follow-up:** Escalate the conflict to authorized business and security decision owners; adjust scope/date rather than silently bypass policy.

AK-E09/F09 — scenario anatomy

Model: Source = authorized employee; stimulus = submits valid request; environment = normal operation at forecast peak; artifact = purchase-decision journey; response = returns correct allowed/blocked/pending

result; measure = at least 99.5% of eligible attempts yield a correct usable response per calendar month. **Typical correction:** “No crash” does not measure correctness. **Follow-up:** Without the clock and denominator, teams can inflate results by ignoring slow/missing attempts.

AK-E10/F10 — availability

Model: Authorized administrator submits approved standard change at normal/peak load. The policy update journey returns the saved correct value/time for at least a **fictional 99% of eligible attempts per month**. Monitor errors and incorrect saves. **Typical correction:** Returning a success screen while storing an old value is a failure. **Follow-up:** A correct policy decline is not an outage for a request/response availability SLI; define the success rule clearly and do not count a wrong decline as correct.

AK-E11/F11 — latency

Model: When a finance user submits a valid lookup, during expected peak, 95% of lookups display usable status/age within **2 seconds**, measured monthly from action to display. Slow or missing results are tracked separately. All numbers are fictional. **Typical correction:** Timing only server response misses screen delay. **Follow-up:** A few very slow waits can be hidden by many quick ones; check distribution/percentiles.

AK-E12/F12 — security

Model positive: An authenticated and authorized admin updates a limit for their organization; it succeeds and can be reviewed. **Negative:** An employee tries to edit the limit; denied without modification, with appropriate evidence. Security/control lead approves the relevant authorization requirement. **Typical correction:** Login does not grant cross-role or cross-organization access. **Follow-up:** No. Check permission for the exact action and business object.

AK-E13/F13 — freshness

Model: Publication lag = **4 minutes** (10:00→10:04). Displayed age at 10:12 = **12 minutes** since that source event (assuming no later update). A missing eligible record has no valid publication lag observation; count it as a **freshness failure / missing record**, not remove it from the denominator. Add source-to-view completeness comparison and explicit status labels. **Typical correction:** A fresh view can miss records or display incorrect data. **Follow-up:** Yes, a recent but incorrect value is fresh and wrong.

AK-E14/F14 — changeability

Model: A small change = adding a predefined optional approval step for a known role. After required business/risk approval, the change becomes effective within a **fictional two business days for 90% of eligible changes per quarter**. Tests verify the new approval route, unrelated policy behavior, and a safe reversal when unexpected rejections rise. **Typical correction:** Approval cannot be excluded merely to improve elapsed time if the business requires it. **Follow-up:** Configuration increases the number of combinations/permissions that may need testing, and a bad value can cause broad disruption.

AK-E15/F15 — ranking

Model for changed client: Top three could be (1) changeability because policy changes are daily; (2) availability because policy editing and purchase responses remain critical; (3) latency for employee decisions. Data freshness may move down if weekly finance review is genuinely adequate, **but this must be validated**. Security remains a mandatory constraint and needs its own acceptance tests. **Typical correction:** Do not conflate priority ranking with permission to violate a control. **Follow-up:** A security improvement can be lower on a discretionary investment list once baseline obligations are met, but mandatory safeguards cannot be overruled by a freshness preference.

AK-E16/F16 — KPI card

Model: Business KPI = finance minutes per eligible exception investigated. Definition = total sampled analyst minutes / number of eligible exception cases; collect two comparable pre-change weeks, segment by complexity, then check four weeks after rollout. **Fictional target:** improve from 20 to 12 minutes per case by 90 days. Owner = finance lead. Technical predictor = eligible records visible within 120 seconds. Guardrail = incorrect status-label rate does not rise. **Typical correction:** Without case mix and baseline, a drop might not be caused by the product. **Follow-up:** More features do not demonstrate less time or better customer outcomes.

AK-E17/F17 — measurements

(a) All three records are within **240 seconds**, so 3/3 = **100%**. **(b)** With a fourth eligible record never visible, success is **3/4 = 75%**; retain it in the denominator and label it missing. **(c)** Example business KPI = weekly minutes finance spends searching for purchase records; guardrail = proportion of incorrect provisional/ verified status labels. **Typical correction:** Filtering away missing rows creates survivorship bias. **Follow-up:** Investigate user workflow, correctness, missing records, process changes, and measurement differences; freshness alone did not prove a business result.

AK-CE01 — model one-page problem brief (fictional)

> **Everything in this example is a [Hypothetical practice assumption] unless it is a statement of what is still unknown. It is not a description of American Express systems.**

Problem: Northstar Office Group says finance sees employee purchases too late to investigate exceptions. Existing delay and case volume are **unknown** and require baseline measurement.

Outcome: Reduce finance time locating a purchase while maintaining correct status labels, appropriate access, and safe risk review.

Customer types and roles: Small organization with one finance generalist and a larger organization with separate finance teams are two possible customer contexts. Administrator sets limits; employee requests purchases; finance locates/checks records; risk reviewer handles exceptions.

Journey: Employee needs supplies → enters request → policy checked → request allowed/blocked/needs review → business result displayed → status recorded/updated → finance finds record → any discrepancy is investigated. **Exception:** A pending or missing status is explicitly labeled and routed to an owner.

Capabilities: Manage policies; evaluate requests; view statuses; investigate exceptions.

Stakeholder and governance view: Product sponsor owns investment outcome; finance lead accepts user workflow and finance visibility target; risk lead owns risk-verification acceptance; security lead reviews access requirements; engineering lead produces feasibility/measurement evidence; operations owns escalation; users validate real tasks. Actual governance is **unknown until confirmed**.

Three example functional requirements:

- `FR-01`: An authorized finance user can search a known purchase ID and see status, timestamp, and provisional/verified label. **Tests:** known record appears; missing record has a defined response; wrong organization cannot see it.
- `FR-02`: An authorized administrator can submit an approved spending-limit change and see effective value/time. **Tests:** approved valid change becomes effective; invalid amount is rejected; unauthorized change is rejected.
- `FR-03`: An employee whose request is blocked sees an appropriate reason and next action. **Tests:** clear response for blocked request; no misleading "approved" label; restricted business details are not disclosed.

Uncertainty log: `CO-01` *proposed constraint*: no unauthorized cross-company data view—confirm exact mandatory policy with security; `CO-02` *proposed constraint*: verified label requires named risk checks—confirm with risk. `AS-01` records can be first displayed within 120 seconds; `AS-02` some finance tasks benefit from provisional records; `AS-03` timestamps can be correlated across capabilities. `OQ-01` what is today's delay? `OQ-02` what exactly is "verified"? `OQ-03` what use cases need faster-than-five-minute visibility? **Owners:** Engineering/data for baseline/time; Finance for user tasks; Risk for label criteria.

Five quality scenarios (fictional numerical proposals; not commitments):

Quality	Trigger / environment / response	Measure and boundary
Availability	Employee submits valid purchase during expected load; usable correct decision returned	At least 99.5% of eligible attempts per calendar month
Latency	Employee submits valid request; correct initial result shown	p95 at most 2 seconds, action-to-display, rolling 28 days
Security	User from organization A attempts to read organization B's record; access denied, no details	All agreed unauthorized automated cases pass, reviewed by security; residual risk assessed
Data freshness	Eligible purchase event recorded; provisional/verified finance row appears with timestamp	At least 99% visible within 120 seconds each month, missing records counted as failures
Changeability	Approved routine spending-limit update; new value safely effective	At least 95% within one business day after final approval per quarter, no unrelated rule regression

Top three among negotiable improvements: (1) Data freshness, because it directly addresses the complaint; (2) availability, because users cannot complete critical tasks when it fails; (3) latency, because purchase decisions need timely responses. **Mandatory security and risk rules remain constraints**, not lower-priority optional work. Changeability is still measured but may be less urgent unless stakeholder research changes this order.

KPI plan: Business KPI 1 = median/typical finance minutes per eligible exception case (baseline not measured; **proposed**, not actual, improvement target of 25% after pilot). KPI 2 = percentage of finance lookup attempts completed without support (baseline not measured; fictional target 90%). Technical signals = eligible records visible within 120 seconds, successful purchase journey ratio. Guardrails = wrong-status labeling and unauthorized data exposure tests. Confirm population, sample sizes, collection method, window, and owners before claiming improvement.

Evidence and decision request: Conduct interviews with finance/employee/admin users; sample actual source-to-visible delays and missing IDs; measure finance effort before rollout; run a controlled pilot of provisional and delayed-confirmed views; review correctness and access. Ask finance/product decision owner to select acceptable visibility behavior after risk/security control review. **Revisit:** if missing records, wrong labels, or finance benefit fail agreed pilot acceptance.

Ten fully worded model clarifying questions

1. What decision does a finance employee make after seeing a new transaction?
2. Which tasks need the information first: employee confirmation, finance review, or exception repair?
3. How long do records currently take to become visible, including the worst cases and missing records?
4. Can finance act on a clearly marked provisional status, or must the status be verified first?
5. Which source event starts the freshness clock, and which screen event stops it?
6. What should a finance user see if an expected record is absent or an earlier status was corrected?
7. Which organization and role boundaries must be enforced on transaction searches?
8. Which risk checks or approvals must finish before a status can be called verified?
9. Which roles approve business timing targets, risk controls, and unresolved exceptions?
10. Which finance time/cost measure will prove that faster visibility created real business value?

Reviewer notes: A good independent submission need not match this ranking or values. It must keep unknowns explicit, name five qualities, define measurements, and explain trade-offs. Do not copy fictional “baselines” into an interview as if they were measured.

AK-PQ — short answer sketches

ID	Core point expected
PQ01	Organization is customer; administrator/employee/finance are users with task-specific roles.
PQ02	Capability = manage policies; process = submit-review-approve; feature = edit-limit screen.
PQ03	Trigger, policy decision, exception branch, status visibility, finance investigation; name handoffs.

ID	Core point expected
PQ04	Security or support is a stakeholder even without being an end user.
PQ05	Finance defines business acceptance, risk/security approves controls, engineering provides feasibility; actual decision rights must be confirmed.
PQ06	Technology request skips discovery of business need and alternatives.
PQ07	Actor+trigger+observable result, plus allowed and denied tests.
PQ08	Unverified assumption with evidence owner, not a confirmed fact.
PQ09	Verification = requirement is clear/correctly specified; validation = it supports a real need.
PQ10	Source, stimulus, environment, artifact, response, response measure.
PQ11	Correct rule-based decline may be successful processing; wrongful decline or no response may be failure.
PQ12	Tail/user experience matters; averages and server-only timings hide bad waits.
PQ13	Own-organization authorized read allowed; other-organization read denied with no disclosure.
PQ14	Source-to-first-visible versus most recent source-update-to-viewing time.
PQ15	Approved small-rule change becomes effective within stated time and without regression.
PQ16	Rank by effect on user and business outcome; keep mandatory controls separate.
PQ17	Finance investigation minutes (KPI) linked to publication-lag ratio (technical measure).
PQ18	Eligible population, clock boundaries, status correctness, thresholds, period, exclusions, missing-data policy, owner.
PQ19	Correctness, adoption, process, training, case mix, or data gaps may offset speed improvement.
PQ20	Define alternatives, quantify trade-offs, validate with finance/risk, and obtain authorized approval; monitor outcomes.

AK-SD01 — sample senior decision defense (fictional)

1. Business goal: "Finance needs timely, trustworthy purchase visibility to investigate exceptions. 'First visible' is separate from 'risk verified' and must not imply final payment completion."

2. Know versus assume: Known from the fictional prompt: finance wants "real time," engineering proposes five minutes, risk requires stronger checks. **Not known:** existing lag, allowable provisional action, risk-check duration, volume, cost, source completeness, and status semantics. **Proposed thresholds below are assumptions.**

3. Options:

Option	Hypothetical user-visible contract	Main benefits	Main costs/risks
A — verified only	At least 99% of eligible records visible **within 5 minutes after risk verification completes**, monthly; show no "verified" record before checks	Simple meaning, avoids premature verified label	Total delay can exceed 5 minutes from purchase; finance may wait
B — early provisional	At least 99% of eligible source records visible as clearly **provisional within 60 seconds**; separately update verified status after checks	Fast awareness; early exception detection	More status handling, user education, risk of misleading use
C — staged pilot	Start with B for internal finance test group; compare outcomes, then expand or fall back to A	Evidence before broad commitment	Additional pilot support and dual workflows

Precision: Option A's five-minute clock starts **after** verification, while B's starts at the agreed **source-event boundary**. They are **not directly comparable as end-to-end delay targets**. For a fair business comparison also measure event-to-first-usable-record and event-to-verified-status across both options. If engineering's actual proposal meant five minutes from the event, verify and correct the contract before deciding.

4. Decision: Recommend **Option C**, conditional on finance confirming provisional rows help its work and risk/security approving safe labels and access. Pilot Option B with explicit "pending verification," while retaining a verified-only view or clear filter for actions that require certainty. If the pilot shows many misleading records or poor benefit, choose verified-only presentation or narrow provisional access. Trade-off: better visibility at extra complexity, not weaker verification.

5. Safety and data: Enforce role/organization access in requirements; make provisional status unmistakable; never silently change historical meaning; show data age; account for missing and late records; specify what users can do while verification is pending; involve risk and security in applicable approval. Report failure counts and a recovery/escalation owner. **Do not claim final financial settlement based on a visible row.**

6. Evidence: Observe a sample of real finance tasks (with approved access), measure current event-to-visible and event-to-verified durations, count missing/wrong-labeled records against the source population,

compare finance time per case, test high-volume and degraded scenarios, and verify that controls work. Define the pilot target and decision date *before* observing results.

7. Ownership: Finance product/sponsor role approves business utility and budget trade-off; risk lead approves risk-verification semantics and permissible actions; security lead approves required access controls; engineering lead owns feasibility and technical evidence; data owner defines eligible source records; operations owns monitoring/escalation; finance lead owns KPI and user acceptance. Exact governance must be confirmed in the actual company.

Fictional outcome targets (not real commitments): Reduce finance minutes per exception by at least 25% relative to an actually collected baseline; record at least 99% of eligible items as visible within 60 seconds during pilot; keep agreed wrong-label and unauthorized-exposure negative tests passing; have zero *uninvestigated* critical missing-record exceptions at pilot review, without claiming zero risk.

Changed-constraint defense:

- **15-minute risk checks for 5% of requests:** Keep them *pending* until verified; report verified-status latency separately. Do not redefine verified or silently exclude slow cases.
- **Finance sees missing transactions:** Stop claiming success from visible rows alone; compare eligible source IDs with view IDs, identify gap, document impact, escalate, and decide rollback/limitation according to pre-agreed rules.
- **30% less capacity:** Protect mandatory controls and measurement. Reduce pilot population or optional features, or choose the five-minute verified-only option if it meets evidenced finance need. Seek sponsor approval for reduced benefits and schedule.

A natural two-minute interview answer:

> "I would first clarify what finance means by real time: seeing a purchase quickly or seeing a fully verified result. Those are different needs. I would compare a delayed verified view with a faster provisional view that clearly shows pending status. I would not weaken required risk checks to hit a display target. My initial recommendation would be a limited pilot, with source-to-visible time, missing records, incorrect statuses, access tests, and finance investigation time measured together. Finance and the product sponsor would approve the business experience and investment trade-off, while risk and security would approve their control requirements. I would revisit the choice if provisional data misleads users or if the benefit is not measurable."

Why this sounds senior: It names the business purpose; distinguishes appearance from correctness; refuses false certainty; compares real options; requests evidence; identifies who accepts which risk; and states a reversal condition. It does not recite technology names or claim personal experience.

Coverage matrix — authoritative checklist ID → lesson → assessment

[Source requirement] Each required item receives an explicit lesson and an independent explanation/application/trade-off check. All five are **Core**; there are **no F01 Depth checklist IDs**. Lesson subdivisions preserve each original item's full meaning rather than silently combining it.

Checklist ID	Required text (preserved from v3 in substance)	Lessons	Specific independent assessments and evidence	Status of guide coverage
C014 · Core	Identify customer types, user roles, business capabilities, and end-to-end journeys	L01 customer types/roles; L02 capabilities; L03 journeys	E01/A01 roles; E02/A02 capability map; E03/A03 full journey + exception; PQ01–PQ03; CE01 customers/capabilities/journey; interview F01–F03	**Taught / exercised / answer-keyed; learner not yet assessed**
C015 · Core	Map stakeholders, decision owners, affected teams, and conflicting objectives	L04 affected teams, stakeholder map, decision roles; L05 conflicts, authority, escalation	E04/A04 impact map; E05/A05 conflict/decision-rights note; PQ04–PQ05; CE01 stakeholder map; SD01 ownership defense; F04–F05	**Taught / exercised / answer-keyed; learner not yet assessed**
C016 · Core	Elicit functional requirements, constraints, assumptions, and open questions	L06 discovery/elicitation; L07 functional requirements/tests; L08 constraints/assumptions/open questions	E06/A06 ten questions; E07/A07 functional requirements/acceptance; E08/A08 classifications; PQ06–PQ09; CE01 requirements and uncertainty log; F06–F08	**Taught / exercised / answer-keyed; learner not yet assessed**
C017 · Core	Write measurable quality-attribute scenarios for availability, latency, security, data freshness, and changeability; rank by business impact	L09 scenario syntax; L10 availability; L11 latency; L12 security; L13 freshness; L14 changeability; L15 ranking	E09/A09 scenario template; E10/A10 availability; E11/A11 latency; E12/A12 security; E13/A13 freshness; E14/A14 changeability; E15/A15 ranking; PQ10–PQ16; five CE01 scenarios; SD01; F09–F15	**Taught / exercised / answer-keyed; learner not yet assessed**

Checklist ID	Required text (preserved from v3 in substance)	Lessons	Specific independent assessments and evidence	Status of guide coverage
C018 · Core	Define business key performance indicators (KPIs) and technical measures that show whether the solution worked	L16 KPI, baseline/ owner/guardrail; L17 calculation and business evidence	E16/A16 KPI card; E17/ A17 metric calculation; Python sample and checks; PQ17–PQ19; CE01 business/technical measures; SD01 evidence; F16–F17	**Taught / exercised / answer-keyed; learner not yet assessed**

Explicit sub-coverage checklist:

- ⊡ C014 customer *types*, user *roles*, business *capabilities*, and end-to-end *journeys*, including exception paths.
- ⊡ C015 stakeholder map, decision owners, affected teams, conflict identification, authority, and escalation.
- ⊡ C016 elicitation questions, functional requirements, constraints, assumptions, unanswered questions, acceptance, verification versus validation.
- ⊡ C017 scenario grammar; **availability**; **latency**; **security**; **data freshness**; **changeability**; ranking based on business impact and hard controls.
- ⊡ C018 outcome-focused business KPIs, technical measures, numerator/denominator, baselines, guardrails, evidence, owners, and evaluation.
- ⊡ Every lesson has purpose/prerequisites, terminology, worked example, guided exercise, failure/trade-off, interview phrasing and follow-up.
- ⊡ Separate answer key for every exercise, cumulative practice, follow-up, and senior challenge.
- ⊡ One cumulative brief, ten questions, quality ranking, senior test, scoring rubric, references, and final audit.

Important: `[x]` above means **the reference document contains the requested teaching/assessment unit**, **not** that the learner has passed it.

Standard scoring rubric, evidence, and completion criteria

Five-level rubric required by topic map v3

Score	Observable evidence
1/5	Cannot explain the concept accurately, even with prompts.
2/5	Knows basic terms, but needs help applying them or misses assumptions/risks.
3/5	Proposes a workable answer, but compares options or failure cases weakly.
4/5	Independently explains, applies, compares realistic choices, and defends a trade-off using checks/evidence.
5/5	Meets 4/5 and adapts under new constraints, identifying second-order business, data, security, operations, and delivery consequences when relevant.

Score each applicable dimension separately

Dimension	What earns at least 4/5 in F01	Evidence location
Business framing	States customer, user journey, business problem and target outcome without jumping to technology	A01–A03, CE01, SD01
Technical correctness	Correctly distinguishes requirement types, five quality scenarios, clocks, populations, and KPI calculations	A06–A17, PQ, SD01
Trade-offs	Names at least two realistic options; explains who gains/loses and changed constraints	A05, A15, SD01
Failure, security, and data risk	Includes exception path, access prohibition, provisional/missing/wrong status, risk/control ownership	A03, A05, A12, A13, SD01
Evidence	Names baseline, sample/data source, acceptance test, metric and uncertainty check	A08–A17, CE01, SD01
Execution and ownership	Assigns decision, data, control, support, and measurement roles; includes review trigger	A04, A05, CE01, SD01

Dimension	What earns at least 4/5 in F01	Evidence location
Communication	Explains decision in simple, correct terms and handles follow-ups without inventing facts	F01–F17, SD01 spoken defense

Completion gate (not automatically achieved)

A learner **passes F01** only when **all** of the following are true:

1. **Every Core checklist item C014–C018 has evidence of three abilities without notes:** explain it accurately, apply it to a new example, **and** defend at least one relevant trade-off or choice. Record a score of **at least 4/5** for each item.
2. **All 17 guided exercises** have been attempted and checked with the answer key or a reviewer. Incorrect answers have been corrected and retested. The learner can explain each corrected result, rather than merely copying it.
3. The **CE01 cumulative practice** contains the one-page brief, **ten fully worded questions**, top-three impact ranking, all five quality scenarios, requirement/assumption/owner tables, business KPIs and technical measures. Its reasoning must independently meet **4/5** on applicable dimensions.
4. **SD01 senior decision test** addresses all seven requested elements, includes at least two quantified options and ownership, responds to the three changed-constraint prompts, and scores **at least 4/5 in every applicable rubric dimension**.
5. The learner can distinguish **verified fact, source requirement, hypothetical assumption, and recommendation** and does not invent employer systems, personal projects, or achievements.
6. Record any remaining gaps, evidence needed, owners and retest dates. **Only then** mark the F01 card complete. **Depth items:** none assigned to F01.

Why the threshold is demanding: The topic map asks for senior judgment, not memorized terminology. A 3/5 answer may be useful practice but is **not** a passing senior answer. No simulated score is presented as an actual learner result.

Blank completion scorecard (fill after independent assessment)

Unit	Current score (1–5)	Explain without notes?	Apply?	Defend trade-off?	Evidence / missed point / retest
C014	Not assessed	—	—	—	
C015	Not assessed	—	—	—	
C016	Not assessed	—	—	—	
C017	Not assessed	—	—	—	
C018	Not assessed	—	—	—	
CE01 cumulative	Not assessed	—	—	—	

Unit	Current score (1–5)	Explain without notes?	Apply?	Defend trade-off?	Evidence / missed point / retest
SD01 business framing	Not assessed	—	—	—	
SD01 technical correctness	Not assessed	—	—	—	
SD01 trade-offs	Not assessed	—	—	—	
SD01 failure/security/data risk	Not assessed	—	—	—	
SD01 evidence	Not assessed	—	—	—	
SD01 execution/ownership	Not assessed	—	—	—	
SD01 communication	Not assessed	—	—	—	

Suggested gap log fields [Recommendation]: date; checklist ID; failed explanation/test; misconception; corrected answer; evidence still needed; next practice scenario; reviewer; retest result. This is a simple study log, not an extra F01 requirement.

Primary references and verification record

Verification date for the external pages below: 2026-10-24. References establish industry/business-analysis terminology and methods. They **do not** establish the fictional Northstar numbers, any implementation at American Express, or the specific interview requirements.

Ref	Issuing body and official primary page	Verified claim and limits
R1	**IIBA**, BABOK Guide glossary	Definitions of stakeholder, actor/end user, assumption, elicitation, capability, business process, functional/non-functional requirement, acceptance criteria, KPI-related metric, requirement verification/validation, RACI. Official glossary; examples in lessons are ours.
R2	**IIBA**, The Business Analysis Standard, section 4.4	Distinction between business, stakeholder, and solution requirements, including functional and non-functional. No claim that a single wording template is mandatory.
R3	**Carnegie Mellon SEI**, Quality Attribute Workshop collection and QAWs Third Edition	Stakeholder-originated quality scenarios, business drivers, refinement, ranking, and trade-offs before choosing architecture. Six-field template is a standard teaching form; not presented as an ISO quotation.
R4	**ISO**, ISO/IEC 25010:2023	Published standard provides an ICT/software product quality model to support specification and evaluation. **Only publicly accessible abstract and metadata verified; not full paid standard.** This guide does not reproduce the standard's complete taxonomy.
R5	**Google SRE team**, Implementing SLOs	SLI/SLO methods; user-relevant availability and latency measures; percentile and pipeline freshness examples. Specific targets in this guide are fictional.
R6	**U.S. National Institute of Standards and Technology (NIST)**, Information security glossary	Confidentiality, integrity, and availability as core information security properties. Does not validate detailed control design for a company.

Ref	Issuing body and official primary page	Verified claim and limits
R7	**American Express**, One AP public product page	Publicly described virtual cards, invoice-backed payments, and ACH/check choices. **Domain context only**; no American Express internal architecture or F01 numerical targets inferred.
R8	**Google SRE team**, Product-Focused Reliability for SRE	Distinguishes more user-oriented, end-to-end product measures from narrower system indicators; recognizes measurement cost/coverage trade-offs.

[Recommendation] Before a real interview, recheck primary pages for changes and obtain the actual job posting if available. Do not read the public One AP examples as requirements for an internal team.

Final completeness and verification audit — as of 2026-10-24

Scope and coverage audit

Check	Result	Evidence / limitation
Correct master map used	**PASS**	v3 authoritative F01 card and learning contract. The user's `B01 (C014–C018)` mismatch is explicitly corrected in scope note.
Original F01 checklist inventory preserved	**PASS**	All five **C014–C018**, all **Core**, no invented Depth items.
Every sub-item taught independently	**PASS**	C014 L01–L03; C015 L04–L05; C016 L06–L08; C017 L09–L15 (five separate quality sections + prioritization); C018 L16–L17.
Six learning steps per lesson	**PASS**	Purpose/prerequisites; terms; step-by-step example; guided exercise; failure/trade-off; interview phrasing/follow-up, L01–L17.
Beginner-first progression	**PASS**	Everyday definitions → users → workflows → authority → requirements → measured qualities → outcome proof.
Practice and solutions separated	**PASS**	E01–E17, CE01, PQ01–PQ20, SD01 precede consolidated AK sections.
Topic-map F01 practice task	**PASS**	CE01 requires one-page brief, ten original clarifying questions, ranked top three attributes with business reasons.
Topic-map F01 senior decision test	**PASS**	SD01 includes 7 required elements, options, approvals, changed conditions, answer model, and seven-dimension scoring.
Official primary-source grounding	**PASS with limitation**	IIBA, SEI, ISO metadata, Google SRE, NIST, optional Amex public page. Full ISO 25010 text **not** independently verified.
Quantitative examples clearly fictional	**PASS**	Targets/costs/volumes/users are marked hypothetical; no actual employer measurements or achievements claimed.

Check	Result	Evidence / limitation
Technology scope unchanged	**PASS**	No F02 component design, D02 payment implementation, T06 security architecture, T07 operational design, or B01 full curriculum.
Code is executable Python, line-explained	**PASS — executed 2026-10-24**	L17 sample compiled and ran with Python; output `[60.0, 210.0, 90.0]` and `66.7%`, and both assertions passed. Code is standard-library only.
Interview statements accurate and natural	**PASS (editorial audit)**	Every lesson has spoken phrasing and plain-English translation; senior example avoids false company claims.
Actual learner mastery	**NOT ASSESSED**	No interactive teaching or scored learner responses were requested. The blank scorecard must remain blank until real evidence exists.
Actual vacancy requirements/internal company stack	**NOT VERIFIED**	Requisition/interview plan/internal data unavailable. No unsupported assertions made.
Actual source-to-visible baseline, risk policy, owner approvals	**NOT VERIFIED**	These are precisely the questions and validation work of F01; practice case is fictional.

Remaining missing or unverified items — explicit list

1. **No F01 checklist content is missing from this guide** according to the authoritative five-ID crosswalk. This is a **document coverage** finding, not a learner competency finding.
2. **Full ISO/IEC 25010:2023 standard text not verified.** Only its official published metadata/abstract support the limited claim cited. No proprietary normative text is required for this F01 learning guide.
3. **Actual job requisition, hiring expectations, internal architecture, and internal policies remain unverified.** They must not be inferred from product marketing or the target-role label.
4. **Northstar practice numbers and constraints are unverified by design.** They are learning assumptions; in an actual engagement, replace them with stakeholder-approved definitions and measured baselines.
5. **Learner's completion is unknown.** No exercises, senior challenge, or interview explanations have been submitted or scored. Every checklist item remains **unpassed until demonstrated**.
6. **Python sample runtime check completed.** The extracted code compiled, ran, printed the documented result, and passed both assertions on 2026-10-24. This does not validate the example as a production monitoring system.

Final scope gate: F01 ends at requirements, trade-off framing, and measurement definitions. The next syllabus card, F02, covers detailed architecture views and decision records. Do **not** treat this guide as a substitute for those later cards.

End of guide.